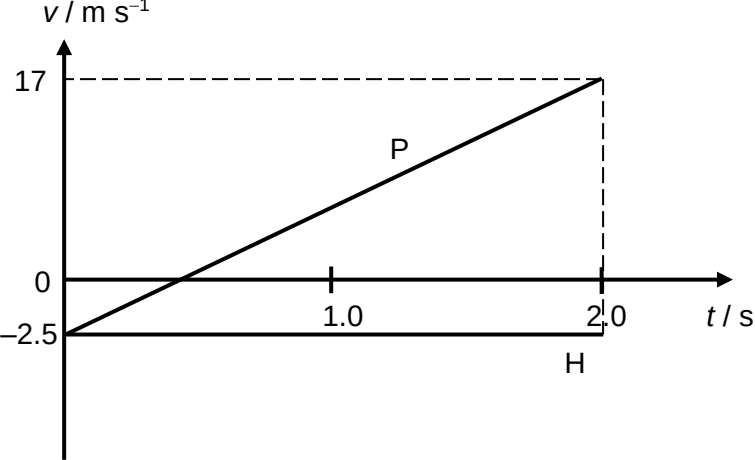


No.	Solution	Remarks
1(a)	$v = u + at = -2.5 + (9.81)(2.0)$ $= 17.1$ Speed is 17 m s^{-1}	[1] substitution [1] answer
1(b)		[1] helicopter: horizontal straight line (-ve value) [1] parcel: diagonal straight line (+ve gradient) (reverse sign accepted) [-1] if not labelled
1(c)	Distance between helicopter and parcel = area of right-angled triangle $= \frac{1}{2}(2.0)(17.1 + 2.5) = 19.6 \text{ m}$ OR From point of release, For helicopter, upwards $s = ut = (2.5)(2.0) = 5.0 \text{ m}$ For parcel, downward $s = ut + \frac{1}{2}at^2$ $= (-2.5)(2.0) + \frac{1}{2}(9.81)(2.0)^2$ $= 14.6 \text{ m}$ Hence their separation $= 5.0 + 14.6 = 19.6 \text{ m}$	[1] working [1] answer
2(a)	The change in height is negligible compared with radius of the	